Frontier of Artificial Network

Department of Data Science
香港城市大學
City University of Hong Kong

An L^2 Convergence Analysis of Neural Approximation via Wavelet Frame Theory



Invited Speaker
Mikyoung Lim
KAIST

Date: November 16, 2026 (Mon)
Time: 14:00 (HKT)
Zoom Meeting: 801 137 0362

Biography

Mikyoung Lim is a professor in the Department of Mathematical Sciences at KAIST. She received her Ph.D. in mathematics from Seoul National University in 2003. She held postdoctoral and faculty positions at École Polytechnique and Colorado State University, respectively, before joining KAIST. Her research interests include inverse problems, spectral analysis of integral operators, and neural network approximation. She was a plenary speaker at the 2025 Applied Inverse Problems (AIP) Conference.

Abstract

We study neural-network approximation with vector-to-scalar activation functions. Using wavelet frame theory on spaces of homogeneous type, we establish an averaging-kernel condition under which the activation function induces a wavelet system. Combining this with L^2-convergence results for wavelet expansions, we obtain convergence estimates for neural approximation, including cases with oscillatory and nonsmooth activation functions.